

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
%pwd
```

'/content'

```
%cd /content/drive/MyDrive/BAIM-660-01 Predictive Text Analytics
```

/content/drive/MyDrive/BAIM-660-01 Predictive Text Analytics

```
%ls
```

AppCodewithDemo.ipynb	combined_data.csv	model.pkl
ASG01_AlleR.ipynb	combined_data.csv.zip	Model_Training.ipynb
ASG02_Rohan_Alle.ipynb	Data_002.csv	non_clickbait_data.csv
ASG03_AlleR.ipynb	Data_002.gsheets	Phase1.ipynb
ASG04_AlleR.ipynb	Data_Cleaning.ipynb	sentiment_analysis.csv
BAIM_660_Project_Main.ipynb	encoder.pkl	spam_samples/
categories.json	imdbDF.csv	tfidf.pkl
clf.pkl	label_encoder.pkl	UpdatedResumeDataSet.csv
clickbait_data.csv	ModelingCode.ipynb	

```
!pip install gradio docx2txt PyPDF2

import gradio as gr
import pickle
import re
import docx2txt
import PyPDF2

# ----- Load your model + vectorizer + encoder -----
tfidf = pickle.load(open("/content/drive/MyDrive/BAIM-660-01 Predictive Text Analytics/tfidf.pkl", "rb"))
model = pickle.load(open("/content/drive/MyDrive/BAIM-660-01 Predictive Text Analytics/model.pkl", "rb"))
le = pickle.load(open("/content/drive/MyDrive/BAIM-660-01 Predictive Text Analytics/label_encoder.pkl", "rb"))

# ----- Resume Text Cleaner -----
def cleanResume(text):
    text = re.sub(r"http\S+|www.\S+", " ", text)
    text = re.sub(r"@S+|#\S+", " ", text)
    text = re.sub(r"^[A-Za-z0-9.+ ]", " ", text)
    text = re.sub(r"\s+", " ", text)
    return text.strip()

# ----- File Extractors -----
def extract_pdf(file):
    reader = PyPDF2.PdfReader(file)
    text = ""
    for page in reader.pages:
        text += page.extract_text() + " "
    return text

def extract_docx(file):
    return docx2txt.process(file)

# ----- Prediction Function -----
def predict_resume(file):
    ext = file.name.split(".")[1].lower()

    if ext == "pdf":
        text = extract_pdf(file)
    elif ext == "docx":
        text = extract_docx(file)
    else:
        text = file.read().decode("utf-8")

    cleaned = cleanResume(text)
    vectorized = tfidf.transform([cleaned])
    pred = model.predict(vectorized)[0]

    # Fix numpy.str_ issue
    if isinstance(pred, str):
        return pred
    else:
        return le.inverse_transform([pred])[0]

# ----- Build Gradio Interface -----
app = gr.Interface(
    fn=predict_resume,
    inputs=gr.File(label="Upload Resume (PDF, DOCX, TXT)"),
    outputs=gr.Label(label="Predicted Job Category"),
    title="AI Resume Screening Tool",
    description="Upload a resume and the model will classify the job category.",
)

# Launch app with public URL
app.launch(share=True)
```

```
Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (5.50.0)
Collecting docx2txt
  Downloading docx2txt-0.9-py3-none-any.whl.metadata (529 bytes)
Collecting PyPDF2
  Downloading pypdf2-3.0.1-py3-none-any.whl.metadata (6.8 kB)
Requirement already satisfied: aiofiles<25.0, >=22.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0, >=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.13.0)
Requirement already satisfied: brotli>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0, >=0.115.2 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.136.3)
Requirement already satisfied: ffmpeg in /usr/local/lib/python3.12/dist-packages (from gradio) (1.0.0)
Requirement already satisfied: gradio-client==1.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.14.0)
Requirement already satisfied: groovy~=0.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: httpx<1.0, >=0.24.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0, >=0.33.5 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.18.0)
Requirement already satisfied: jinja2<4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0, >=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.0.3)
Requirement already satisfied: numpy<3.0, >=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.0.2)
Requirement already satisfied: orjson~=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (3.11.9)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from gradio) (26.2)
Requirement already satisfied: pandas<3.0, >=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.2.2)
Requirement already satisfied: pillow<12.0, >=8.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (11.3.0)
Requirement already satisfied: pydantic<=2.12.3, >=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.12.3)
Requirement already satisfied: pydub in /usr/local/lib/python3.12/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.0.32)
Requirement already satisfied: pyyaml<7.0, >=5.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (6.0.3)
Requirement already satisfied: ruff>=0.9.3 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.15.16)
Requirement already satisfied: safehttpx<0.2.0, >=0.1.6 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<1.0, >=0.40.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.52.1)
Requirement already satisfied: tomkit<0.14.0, >=0.12.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0, >=0.12 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.49.0)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0->gradio) (2025.3.0)
Requirement already satisfied: websockets<16.0, >=13.0 in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0->gradio) (15.0.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0, >=3.0->gradio) (3.18)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0, >=0.115.2->gradio) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0, >=0.115.2->gradio) (0.0.4)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0, >=0.24.1->gradio) (2026.5.20)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0, >=0.24.1->gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0, >=0.24.1->gradio) (0.16.0)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0, >=0.33.5->gradio) (8.4.1)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0, >=0.33.5->gradio) (3.29.2)
Requirement already satisfied: hf-xet<2.0.0, >=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0, >=0.33.5->gradio) (1.5.1)
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0, >=0.33.5->gradio) (4.67.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0, >=1.0->gradio) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0, >=1.0->gradio) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0, >=1.0->gradio) (2026.2)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3, >=2.0->gradio) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3, >=2.0->gradio) (2.41.4)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0, >=0.12->gradio) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0, >=0.12->gradio) (13.9.4)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3.0, >=1.0->gradio) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<1.0, >=0.12->gradio) (4.2.0)
Requirement already satisfied: pygments<3.0.0, >=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<1.0, >=0.12->gradio) (2.20)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer<1.0, >=0.12->gra
Downloading docx2txt-0.9-py3-none-any.whl (4.0 kB)
Downloading pypdf2-3.0.1-py3-none-any.whl (232 kB)
232.6/232.6 kB 5.5 MB/s eta 0:00:00

Installing collected packages: docx2txt, PyPDF2
Successfully installed PyPDF2-3.0.1 docx2txt-0.9
Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://edbd79669ebf40ad89.gradio.live
```

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to dep

AI Resume Screening Tool

Upload a resume and the model will classify the job category.

Upload Resume (PDF, DOCX, TXT)

Resume_May30_2026.pdf79.8 KB

Clear

Submit

Predicted Job Category

Data Science

Flag

Next steps: [Deploy to Cloud Run](#)

MAYBE A COURSE SCREENER FOR WNE?

SCOPE

- ◆ Polish the UI (modern styling, colors, layout)
- ◆ Add Top-3 job predictions
- ◆ Add confidence scores

- ◆ Add resume keyword extraction
- ◆ Add job-matching (resume → job descriptions)